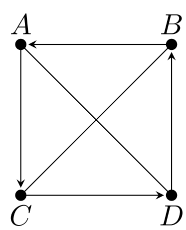
LECTURE 7

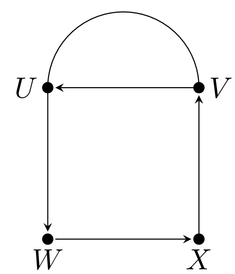
5.5. **Hamilton paths and cycles.**

Definition 5.30. If $G = (V, E)$ is a (multi)graph we say that G has a *Hamilton cycle* if there is a cycle containing each vertex in V. If it is a path instead of a cycle, it is called a *Hamilton path*.

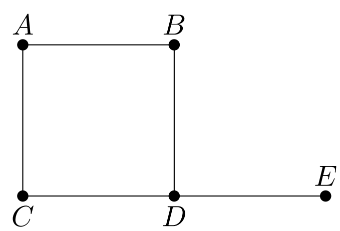
Remember that a cycle never has repeated vertices, so a Hamilton cycle visits every vertex exactly once.

Example 5.31. The following two graphs have a Hamilton cycle as depicted below:





However the following graph does not have a Hamilton cycle, but only a Hamilton path:



Some facts we can use when searching for a Hamilton cycle in a graph $G = (V, E)$:

(i) G must be connected;
(ii) for each $v \in V$, we have $\deg(v) \geq 2$;
(iii) if a vertex $v \in V$ has been included, no more edges from that vertex can be in the Hamilton cycle;
(iv) the Hamilton cycle can never contain smaller cycles.

In some types of graphs we know there is a Hamilton cycle: to see this, we first consider Hamilton paths.

Theorem 5.32. *If $G = (V, E)$ is loop-free with $|V| = n \geq 2$ and $\deg(x) + \deg(y) \geq n - 1$ for each pair $x, y \in V$, with $x \neq y$, then G has a Hamilton path.*

Proof. We first show that G must be connected: suppose C_1, C_2 are two components with $x \in C_1$, $y \in C_2$, and for $i \in \{1, 2\}$, let n_i be the number of vertices in C_i. Then $\deg(x) + \deg(y) \leq (n_1 - 1) + (n_2 - 1) < n - 1$, a contradiction.

We now build a Hamilton path: start with any path $P_m : v_1 \to v_2 \to \cdots \to v_m$, for some $m \geq 2$. (For example, an edge with its endpoints.) If possible, add vertices to make the path longer both at the beginning and at the end. Repeat this until this new path $P_k : v_1 \to v_2 \to \cdots \to v_k$ has v_1 and v_k adjacent only to vertices in the path P_k. Look at $N(v_1)$, which is the set of vertices adjacent to v_1. We want to show that if $k < n$ we can prolong the path.

First we claim that the vertices $v_1, v_2, \ldots, v_k$ lie on a cycle of length k. Indeed, either $v_1 \sim v_k$ (and we have a cycle $v_1 \to v_2 \to \cdots \to v_k \to v_1$), or $v_1 \not\sim v_k$. In the latter case, recall that v_1 and v_k are adjacent only to vertices in P_k. In particular v_1 is adjacent to a subset of vertices in $\{v_2, \ldots, v_{k-1}\}$. Note that if $v_1 \sim v_t$ and $v_k \sim v_{t-1}$, for some $t \in \{3, \ldots, k-1\}$, then we have a cycle $v_1 \to v_2 \to \cdots \to v_{t-1} \to v_k \to v_{k-1} \cdots \to v_t \to v_1$. Hence we may assume that whenever $v_1 \sim v_t$ for some $t \in \{2, \ldots, k-1\}$, we have $v_k \not\sim v_{t-1}$. Suppose that v_1 is connected to m vertices in $\{v_2, \ldots, v_{k-1}\}$. It then follows that $\deg(v_k) \leq (k-1) - m$. Thus,

$$\deg(v_1) + \deg(v_k) = m + \deg(v_k) \leq k - 1 < n - 1,$$

which contradictions our assumption. Hence the vertices $v_1, v_2, \ldots, v_k$ lie on a cycle of length k, as required.

Now, since we are assuming that $k < n$, there is a vertex $v \in V$ which is not on this cycle. Since G is connected, there is a path from v to a vertex v_r, for $r \in \{1, \ldots, k\}$. Removing the edge $\{v_{r-1}, v_r\}$ (or $\{v_1, v_r\}$ if $r = t$ as above), we get a longer path. Repeating this process, we can continue to increase the length of the path until it includes every vertex in V. $\square$

Note that there are many graphs, not satisfying the conditions of the above theorem, but still with a Hamilton path. For example, graphs of the form:

Corollary 5.33. *If $G = (V, E)$ is loop-free and $|V| = n \geq 2$ and each vertex has degree $\geq \frac{n-1}{2}$ then G has a Hamilton path.*

Proof. This is immediate from the theorem since $\frac{n-1}{2} + \frac{n-1}{2} = n - 1$. $\square$

Now to Hamilton cycles.

Theorem 5.34. *If $G = (V, E)$ is loop-free and undirected with $|V| = n \geq 3$ and $\deg(x) + \deg(y) \geq n$ for all non-adjacent $x, y \in V$, then G contains a Hamilton cycle.*

Proof. If G has no Hamilton cycle, add edges until you get a graph H where any additional edge would create a Hamilton cycle. Now add some edge $\{a, b\}$ to H. Suppose we have the Hamilton cycle:

$$a \to b \to v_1 \to v_2 \to \cdots \to v_n \to a$$

If $b \sim v_i$ then $a \not\sim v_{i-1}$ as otherwise we have a Hamilton cycle without $a \to b$ in H; compare with the proof of the previous theorem. Thus every link to b prohibits a link from a. Therefore

$$\deg_H(a) + \deg_H(b) \leq n - 1 < n.$$

In G the degrees are even smaller so

$$\deg(a) + \deg(b) < n \qquad \text{and} \qquad a \not\sim b,$$

which yields the required contradiction. $\square$

Corollary 5.35. *If $\deg(v) \geq \frac{n}{2}$ for all $v \in V$ then G has a Hamilton cycle.*

Proof. Immediate from Theorem 5.34 since $\frac{n}{2} + \frac{n}{2} = n$. $\square$

Corollary 5.36. *If $|E| \geq \binom{n-1}{2} + 2$ then G has a Hamilton cycle.*

Proof. Let $a, b \in V$ be such that $a \not\sim b$. Remove a, b and their edges and call the resulting graph $H = (V', E')$. Note that H is a subgraph of the complete graph K_{n-2} since $|V'| = n-2$. Thus $|E'| \leq \binom{n-2}{2}$, and we obtain

$$\binom{n-1}{2} + 2 \leq |E| = |E'| + \deg(a) + \deg(b) \leq \binom{n-2}{2} + \deg(a) + \deg(b).$$

This implies that

$$\deg(a) + \deg(b) \geq \binom{n-1}{2} + 2 - \binom{n-2}{2} = n.$$

[The equality above can be checked via a direct computation, or using the identity $\binom{m}{k} = \binom{m-1}{k} + \binom{m-1}{k-1}$.] So Theorem 5.34 applies. $\square$

Remark. We have $\binom{n}{2} - \left[\binom{n-1}{2} + 2\right] = n-3$, so for the corollary to apply we may only remove at most $n-3$ edges from the complete graph. For example, we can remove up to 3 edges from K_6.

Definition 5.37. A complete directed graph on n vertices is denoted by K_n^* and defined as followed: for each distinct vertices x, y, exactly one of the edges (x, y) or (y, x) is in K_n^*. Such a graph is called a *tournament*.

Theorem 5.38. *Every tournament has a directed Hamilton path.*

Proof. Consider a tournament K_n^*, and for $m \geq 2$, consider a path $P_m : v_1 \to v_2 \to \cdots \to v_m$. If $m = n$, we are done. Hence suppose $m < n$, and let v be a vertex that does not appear in P_m.

If (v, v_1) is a directed edge in K_n^*, we can extend P_m by adjoining this edge. If (v, v_1) is not an edge in K_n^*, by the definition of a tournament, we have that (v_1, v) is an edge in K_n^*. Next suppose (v, v_2) is in K_n^*. Likewise we have the larger path

$$v_1 \to v \to v_2 \to \cdots \to v_m.$$

If (v, v_2) is not in K_n^*, then (v_2, v) is in K_n^*. As we continue this process, there are only two possibilities:

(a) for some $1 \leq k \leq m-1$, the edges (v_k, v) and (v, v_{k+1}) are in K_n^*, and in this case we replace (v_k, v_{k+1}) with this pair of edges; or
(b) (v_m, v) is in K_n^*, and we then add this edge to the path P_m.

Either case results in a longer path P_{m+1}. We then repeat this process until we reach a path P_n on the n vertices. $\square$